

CORESTAR ANALYSIS

SETUP GUIDE

Six steps to build a setup you can trust before you look at a single tube.

1 LOAD THE DATA

Open the exam data and the calibration file. Confirm the probe type, frequencies, and tube range match the inspection plan.

CHECK Cal standard ID matches the data sheet



2 SET UP CHANNELS

Assign differential and absolute channels for each frequency. Build any required mixes (for example, support suppression) per your procedure.

CHECK Every channel is labeled and in the right order



CAL STD

3 SET THE PHASE

On the cal standard, rotate so probe motion is horizontal. Set the through-wall hole signal to the angle your procedure calls for.

CHECK Signal travels in the expected direction

4 SET THE VOLTAGE

Normalize the reference signal (for example, the 4 × 20% flat-bottom holes) to the voltage in your procedure. Apply it to all channels.

CHECK Reference reads the target voltage

5 BUILD THE CURVE

Mark each cal flaw (100%, 60%, 40%, 20%) and save the phase-to-depth curve. Points should form a smooth line.

CHECK No point sits far off the curve



6 SAVE & VERIFY

Save the setup with a clear name. Open a known tube and confirm supports and landmarks look normal before you start.

CHECK Setup name includes unit, probe & date



QUICK TIPS

Change one setting at a time and re-check the cal standard after each change.

Use the same setup for every tube in the same exam group.

If a step won't line up, stop and ask. Don't force it.

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HOW TO ANALYZE

A repeatable routine for every tube: scan, spot, measure, size, code, and report.

1 SCAN THE TUBE

Scroll the strip chart along the full length of the tube. Find the landmarks first: tube sheet, supports, and U-bend.

CHECK All expected landmarks are present



2 SPOT SIGNALS

Look for anything that breaks the normal background or looks different from a typical support signal.

CHECK Review the primary channel first, then the others



3 READ THE LISSAJOUS

Box the signal and look at its shape in the Lissajous window. Measure the peak-to-peak voltage and note the phase angle.

CHECK Signal is clean, not noise or probe wobble

4 SIZE IT

Read the depth from your calibration curve using the phase angle. Confirm the call on a second frequency or mix.

CHECK Both frequencies tell the same story

5 CODE & LOCATE

Give it the right report code (for example, % depth, DNT, or NDD). Record the location in inches from the nearest landmark.

CHECK Code comes from the approved code list



6 REPORT & REVIEW

Enter the result, save, and move to the next tube. Flag anything unusual for a second analyst or resolution review.

CHECK Every tube has a result, even if it's NDD



QUICK TIPS

When unsure, compare the signal to the same spot on a clean tube.

Always measure location from the nearest landmark, not the tube end.

If you can't call it with confidence, flag it. Don't guess.